

The Language of the Run Game: Gaps, Techniques, and Blocks

The words coaches use for the spaces in a line, where every defender stands, and what every blocker does, and why each word exists

The replay rolls, and the analyst draws on the screen. “Look at this. They’ve got a 3-technique over the right guard, and they run right at him. The guard and tackle double-team him, the tackle climbs to the Mike, the fullback kicks out the end, and the back hits the C gap untouched.” You nodded, because he sounded certain. But almost every noun in that sentence was a word nobody ever taught you: 3-technique, double team, climbs, Mike, kicks out, the C gap. This chapter teaches them. It is the vocabulary the rest of the run game is written in, and it is smaller than it sounds: a handful of names for *spaces*, a numbering for *where a defender stands*, and eight verbs for *what a blocker does*. Each one exists because it solves a problem. Learn the problems and the words stick.

You’ll be able to:

- Name every gap from A to D and read a numbered run call (“24”) on a diagram.
- Look at a defensive line before the snap and describe it in technique numbers (“5, 1, 3, 9”), and say which offensive linemen are covered and which are uncovered.
- Recognize the eight core blocks (drive, down, reach, cut-off, combo, pull, kick-out and lead) and say what each is for.
- Count the box and do the “hat math”: who is left unblocked, and why the defense can almost always arrange to have one more man than the offense can block.
- Explain why a quarterback who only hands off leaves the offense a defender short, and how a quarterback who can run gets that man back.
- Say in two sentences how zone and gap blocking differ, and what each one trades away.

How to read the diagrams. Offense is blue circles, defense orange squares. *Left* and *right* always mean the offense’s left and right, and plays are drawn going up the page. LT, LG, C, RG and RT are the five offensive linemen (in the small block diagrams, just T, G and C); Q is the quarterback, R the running back, F a fullback, and Y and U are tight ends on the right and left. Defensive linemen are labelled with their *technique number*, which this chapter teaches; M, W and S are the Mike, Will and Sam linebackers you met in The Twenty-Two. Lines ending in a short bar are blocks; arrows are paths; the thick dark arrow is the ball carrier.

Why the run game needs its own words

A running play lasts about three seconds and involves twenty-two people colliding in a space about fifteen yards wide. The people inside it have to talk about it in two very different moments. Before the snap, a center has maybe ten seconds to tell four other linemen who is blocking whom against a defense that is still moving. Later, in the film room, a line coach has to explain why a play that gained twelve yards last week lost two today. Both conversations need words that are short, unambiguous, and shared by everyone in the room.

So football built a vocabulary for exactly three things:

1. **The spaces** in the line where a runner can go (gaps and holes).
2. **Where each defender stands** relative to the blockers (technique numbers).
3. **What each blocker does** (a short list of named blocks).

Nearly every sentence about the run game, from a high-school coach’s whistle to a network analyst’s telestrator, is built from those three sets of words. One more idea glues them together: *counting*. Run blocking is in the end an arithmetic problem, blockers against defenders, and the last third of this chapter is about that arithmetic.

Gaps: the spaces between blockers

Start with the line itself. When an offense breaks the huddle, its five linemen stand side by side, a little more than an arm’s length apart (the space between them is the lineman’s *split*). The spaces between them, and the space outside the last man, are where a running back can go. Each space has a letter.

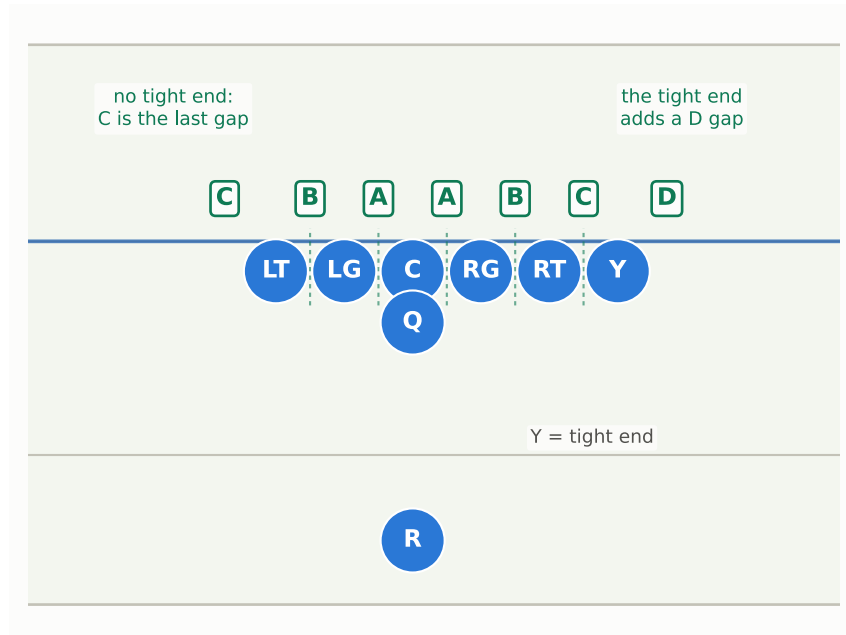


Figure 1: Gaps are lettered outward from the center, the same on both sides: A between the center and a guard, B between a guard and a tackle, C outside a tackle. A tight end (Y) on the line adds one more gap outside himself, the D gap, on his side only. With no tight end on the left, that side ends at the C gap.

A **gap** is the space between two adjacent players on the line of scrimmage, named by a letter counted outward from the center:

- The **A gaps** are on either side of the center, between the center and each guard.
- The **B gaps** are between each guard and the tackle beside him.
- The **C gaps** are just outside each tackle (between the tackle and a tight end, if there is one).
- The **D gap** is outside a tight end. It exists only on a side with a tight end on the line.

There is one A, one B and one C gap on each side, so an offense with no tight end has six gaps, and every tight end on the line adds one more. Some coaches keep going (an “E gap” outside a second tight end or a wing), but A to D covers almost everything you will hear on a broadcast.

Why letters, and why from the defense’s point of view? Gap letters are mostly the *defense’s* language. A run defense’s first job is to make sure that every gap has a defender responsible for it, so that wherever the ball carrier goes, somebody is already there. Coaches call that a defender’s *gap responsibility*: “you have the B gap.” Letters are fast to shout and the same on both sides of the ball, which is why a linebacker can say “A gap, A gap!” and every teammate knows he means the space next to the center. (How defenses divide up the gaps, and what happens when a defender leaves his, is Run Fits and Box Math.)

That point leads straight to the first piece of run-game arithmetic. **Every player the offense puts on the line creates a gap the defense has to fill.** Add a tight end and there are seven gaps instead of six; add a second tight end and there are eight. A defense that wants a defender for every gap needs one more defender near the ball for every blocker the offense attaches to its line. That is a big part of why offenses use tight ends in the run game at all, and why the personnel an offense sends on the field changes how many defenders the defense has to keep close.

🔥 Common misconception

“The A gap is a spot on the field.” It is not. A gap is defined by the *players* on either side of it, so it moves when they move. If the center and right guard both step two yards to their right at the snap, the right A gap moves two yards to the right with them. That sounds pedantic, but it matters as soon as you watch a zone run in 03-02, where the whole line moves sideways together and the gaps slide along the line with it. Gaps also stretch and shrink: a lineman who takes a wider split before the snap makes the gaps on either side of him wider, and defenses watch splits for exactly that reason.

Holes: the offense's numbers

The offense has its own way of naming the same spaces, older than the gap letters and built for a different job: calling a play in the huddle.

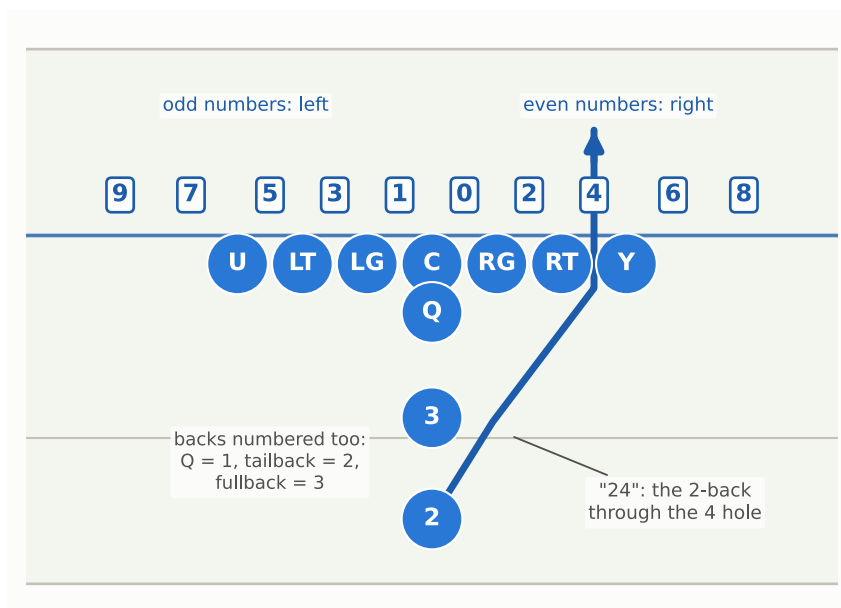


Figure 2: One common hole-numbering system, drawn with two tight ends (U and Y) so both sides match: counting outward from the center, the holes on the right get even numbers (0, 2, 4, 6, 8) and the holes on the left get odd numbers (1, 3, 5, 7, 9). The backs have numbers too. A two-digit call names the ball carrier, then the hole: '24' sends the tailback (the 2-back) through the 4 hole, just outside the right tackle. Many teams flip odd and even, or number the linemen instead of the spaces.

In **hole numbering**, every place a run can go gets a digit. In the most common version, the holes are counted outward from the center with even numbers on one side and odd numbers on the other (in Figure 2, even to the right). The backs get digits too: in one classic scheme, built for the four backs of the old T-formation, the quarterback is 1, the fullback 3, and the halfbacks 2 and 4 (2 on the left, 4 on the right; in an I-formation the tailback is the 2).¹ Put the two together and a run call is just two digits: **"24" means the 2-back carries the ball through the 4 hole**. A coach can add a word for the blocking scheme ("24 Power," "32 Trap") and the whole play fits in one breath.

Why it exists. Before radio helmets and wristbands, a play had to be called in a huddle in a few seconds and remembered by eleven people. A numbering system lets one short code cover hundreds of plays: any back, any hole, any blocking scheme. Linemen learn *rules* for each scheme ("on 24 Power, the play-side tackle blocks down"), so the number tells them which rule applies and which way to apply it.

The catch. Numbering is a dialect, not a standard. Some teams number the *gaps*, as in the picture; others number the *linemen*, so that the "0 hole" is the center himself and the "2 hole" is wherever the right guard was. Some put the odd numbers on the right. And many modern offenses barely use hole numbers in their run calls at all, because their most common runs do not *have* one hole. On a zone run, as you will see in 03-02, the line steps together and the runner picks the hole after the snap; a call like "outside zone right" says where the play *starts*, not where it ends. That is why you will hear NFL coaches name runs by scheme and direction far more often than by number. (How whole playbooks name things is in *Offensive Systems and Their Languages*.)

i Madden vs. real life

Madden's run menu is full of "HB Dive" and "FB Dive" plays aimed at one numbered hole, and a lane lights up when you snap it. Real NFL runs are mostly called by blocking scheme, and the hole on half of them is decided by the defense's reaction after the snap. A hole number tells you where the play is *designed* to go. Whether it goes there is up to eleven defenders.

¹Hole and back numbering vary by team. The even-right, odd-left convention and the "back digit + hole digit" call are described in many coaching primers, e.g. "Gap (American football)," Wikipedia ([https://en.wikipedia.org/wiki/Gap_\(American_football\)](https://en.wikipedia.org/wiki/Gap_(American_football))), and the play-calling primer at <https://s3.amazonaws.com/footballtutorials/concepts/playcalling.pdf> (Football-Tutorials.com: "The QB is always '1', and the FB is always '3', the tailback is 2, and in four-back sets 2 is the back left of the fullback and 4 the back right of him; holes 'numbered odd on the left and even on the right'"). That primer numbers the hole behind the center 0, a reminder that numbering is a dialect; the back numbers are one common scheme, not a standard.

Play side, backside, and the point of attack

Three more words describe direction, and you need them before the blocks make sense.

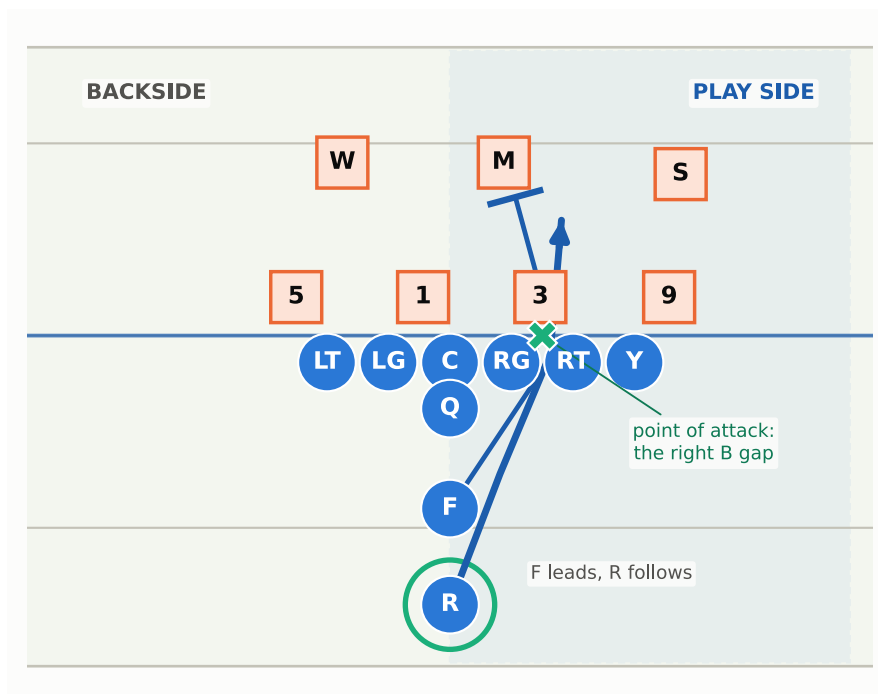


Figure 3: A lead run to the right. The shaded half of the formation is the play side, the half the run is aimed at; the other half is the backside. The point of attack (green X) is the spot on the line where the play is designed to hit: here, the right B gap, between the right guard and right tackle. The fullback (F) leads through it onto the Mike linebacker; the running back follows him.

The **play side** (often written *playside*) is the half of the formation the run is aimed at; the **backside** is the other half, away from the play. The **point of attack** is the exact spot on the line where the play is designed to hit: in Figure 3, the right B gap. Broadcasters abbreviate it “the POA” or say “the point.”

Do not confuse the play side with the *strong side*. The strong side is a fact about the formation (the side with the tight end, or with more players), fixed before the snap. The play side is a fact about the play, and offenses run away from their strong side all the time, precisely because defenses load up toward it.

Why the distinction matters: blockers have completely different jobs on the two sides. On the play side their job is to *make* the hole: move defenders, wall them off, create space at the point of attack. On the backside their job is to stop the defenders on that side from *chasing* the play down from behind. Those backside defenders do not need to be knocked back; they only need to be kept out of the way for a second or two. As you are about to see, there are different blocks for each job, and the side a block is on tells you most of what it is trying to do.

Technique numbers: where the defender stands

Now turn to the defense. Before an offensive lineman can block anyone, he needs to know exactly where the defender in front of him is standing, because a defender on his left shoulder and one on his right shoulder are different problems. Saying “the tackle’s a bit outside the guard” is too slow and too vague. So football gave every spot along the line a number.

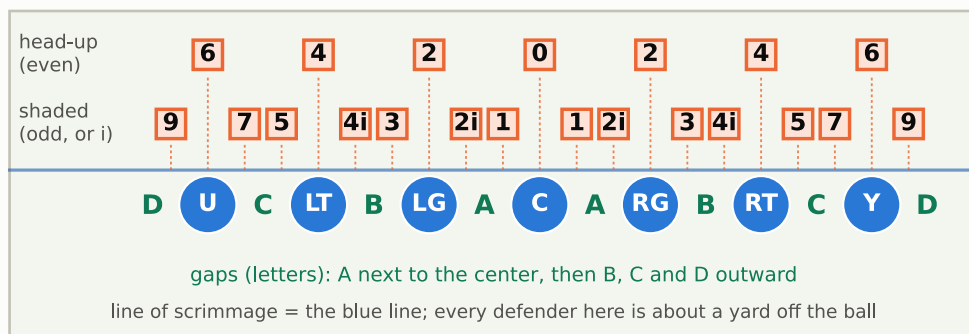


Figure 4: The technique chart. The offensive line is drawn with extra-wide spacing so every number has room; in a real game all these alignments are crammed into the 15 or so yards between the tight ends, about a yard off the ball. Head-up alignments (top row) take even numbers: 0 on the center, 2 on a guard, 4 on a tackle, 6 on a tight end. Shaded alignments (lower row) take odd numbers on a lineman’s outside shoulder (1, 3, 5, 9) or an ‘i’ for an inside shoulder (2i, 4i); the inside shoulder of a tight end is a 7. Green letters are the gaps from earlier: each gap holds one or two shaded alignments.

A **technique** (or *alignment number*) says where a defensive lineman lines up relative to the offensive linemen. The system rests on two ideas:

- A defender lined up directly across from a blocker, nose to nose, is **head-up** on him, and gets an **even** number: **0** head-up on the center, **2** on a guard, **4** on a tackle, **6** on a tight end.
- A defender on one of a blocker’s shoulders is in a **shade**. On the **outside** shoulder he gets the **odd** number after that blocker’s even one: **1** on the center’s shoulder, **3** on the guard’s outside shoulder, **5** on the tackle’s. On an **inside** shoulder he gets the blocker’s even number plus an “i”: **2i** on the guard’s inside shoulder, **4i** on the tackle’s.

The numbers are mirrored: a 3-technique can be on the left guard or the right guard, and the caption or the context tells you which. The pattern breaks at the tight end: in the most common version, a defender on the tight end’s *inside* shoulder is a 7 (not a “6i”), and anything outside him is a 9. Systems disagree here more than anywhere else. Some swap the 6 and the 7, some use an 8 when two tight ends line up on the same side, and many use “wide 9” for an end lined up well outside the tight end.²

Table 1: Where each shaded alignment lives. Outside a tight end, everything is a 9 or wider.

Gap	Alignments that sit in it	Typical player
A	1, 2i (and the 0 covers both A gaps)	nose tackle, “shade” nose
B	3, 4i	3-technique tackle; 3-4 defensive end
C	5, 7	defensive end
D	9	defensive end or outside linebacker on the edge

That table is the reason the system exists. A **shaded defender is usually responsible for the gap he is shaded into**. A 3-technique sits in the B gap and, in most defenses, owns it; a 1-technique owns the A gap he is sitting in. So the alignment number is not just a location: it is a strong hint about the defender’s job. Head-up defenders are different. A 0 or a 4 is often asked to control the blocker in front of him and then fill whichever gap the ball goes to, on either side. That style is called *two-gapping*, and it is taught in Defensive Line Play.

Where the numbers came from. The system is usually credited to Paul “Bear” Bryant, but the better-documented story gives it to **Bum Phillips**, who, by his hometown paper’s account, devised it while coaching at Nederland High School in Texas in the early 1950s, and who joined Bryant’s staff at Texas A&M in 1957. Bryant himself credited Phillips. Before it, every staff used its own code words for alignments, so a lineman who changed schools had to learn a new language.³ Phillips, the Stetson-wearing

²The head-up/shade logic and the variations at the tight end: “Gap (American football),” Wikipedia (7 and 9 “are not universal”): [https://en.wikipedia.org/wiki/Gap_\(American_football\)](https://en.wikipedia.org/wiki/Gap_(American_football)); Big Blue View, “Defensive line techniques, gaps explained” (May 24, 2023: 6 head-up on the tight end, 7 on his inside shoulder as “the only anomaly,” 9 outside him, 8 with two tight ends to one side): <https://bigblueview.com/2023/5/24/23734677/defensive-line-techniques-gaps-explained-big-blue-view-glossary-of-terms>; Scouting Academy, “ITP Glossary: 7 Technique” (the Bryant numbering “swaps the order of the 6 and 7”): <https://scoutingacademy.com/itp-glossary-7-technique/>; Pro Football Focus (Sam Monson, 2011), “Defensive Line Techniques: The Prototypes”: <https://www.pff.com/news/defensive-line-techniques-the-prototypes>.

³Gregg Rosenthal, “Wade Phillips brings quick-fix wizardry to Los Angeles Rams’ D,” NFL.com (2017): Bryant “credited Bum with creating the universally used numbering system which tells defensive linemen where to line up based on their assigned numbers, like the ‘1 technique’ or ‘5 technique,’”

Texan who later coached the Houston Oilers, liked things simple, and the numbers stuck because they *are* simple: two rules, ten digits and an “i”, and a defensive coordinator can describe an entire line in a breath.

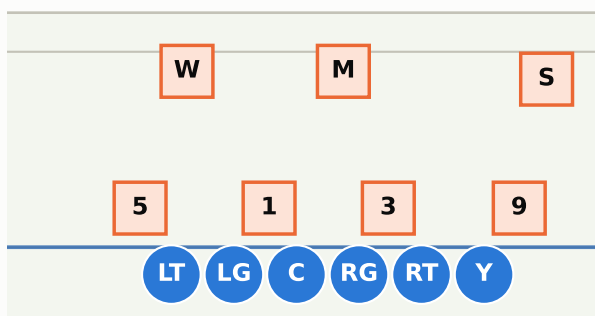
The numbers that became position names

A few technique numbers are used so often that they turned into names for *positions*:

- The **3-technique** is the defensive tackle on a guard’s outside shoulder. In a four-man line he is usually the best pass rusher on the inside, because his alignment puts him one-on-one with a guard in the B gap, with room to shoot past him. When analysts call a player “a 3-technique” they mean that job and that body type: quick, explosive, penetrating. Aaron Donald, whom you met in The Twenty-Two, was the archetype.
- The **5-technique** is on the tackle’s outside shoulder. In a 3-4 defense it became the name for the defensive end: a big, long player who can hold off the tackle and still close the C gap.
- The **4i-technique** sits on the tackle’s *inside* shoulder, in the B gap. It is a favorite alignment for the ends of modern three-man fronts, because a defender there clogs the B gap and makes the offense’s favorite double team, guard and tackle together, awkward to set up.
- The **0** and **1** are the nose. A 0 is head-up on the center; a 1, often called a *shade nose*, sits on one of the center’s shoulders. Both are usually the heaviest men on the field (the nose tackle of 01-03).

Once you know the numbers, a whole front becomes a string you can say out loud.

A four-man front: 5, 1, 3, 9



A five-man front: 5, 4i, 0, 4i, 9

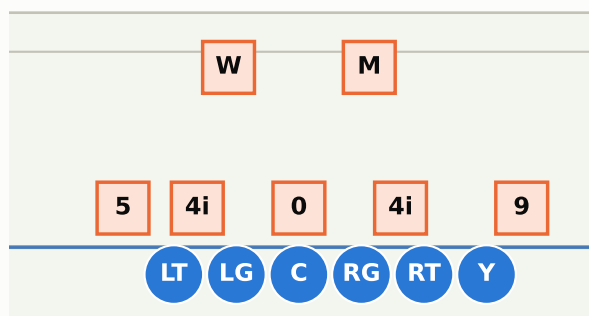


Figure 5: Two fronts, read as numbers from left to right. Left: four linemen in a 5, a 1 (shaded on the center’s left shoulder), a 3 (on the right guard’s outside shoulder) and a 9 (outside the tight end), with three linebackers behind them. Right: five men on the line in a 5, a 4i, a 0, a 4i and a 9, with two linebackers. The names and logic of these fronts are taught in Part 5; here they are just practice in reading the numbers.

Read Figure 5 from left to right and you can describe each front without a single name: “five, one, three, nine” and “five, four-i, zero, four-i, nine.” Coaches do exactly that. The right-hand front has a name, the *tite* front, and a story (why so many NFL defenses adopted it to stop inside runs) that belongs to Odd and Hybrid Fronts. The four-man front on the left is one of the even fronts in 05-02. You do not need either name yet. You need the numbers, because the numbers are what the offensive line reads.

Covered and uncovered linemen

Here is the first thing an offensive lineman does with those numbers. Before every snap, each lineman asks a one-word question: *am I covered?*

which replaced code words that “varied from team to team”: <https://www.nfl.com/news/wade-phillips-brings-quick-fix-wizardry-to-los-angeles-rams-d-0ap3000000834089>. Port Arthur News, “An ode to Bum” (Phillips “while at NHS” taught the numbered techniques to Bryant at Texas A&M): <https://www.panews.com/?p=10285>. Phillips coached Nederland High School 1951–56 and was on Bryant’s Texas A&M staff in 1957: https://en.wikipedia.org/wiki/Bum_Phillips. (NFL.com dates the A&M hire to 1958, but Bryant left A&M for Alabama after the 1957 season.)

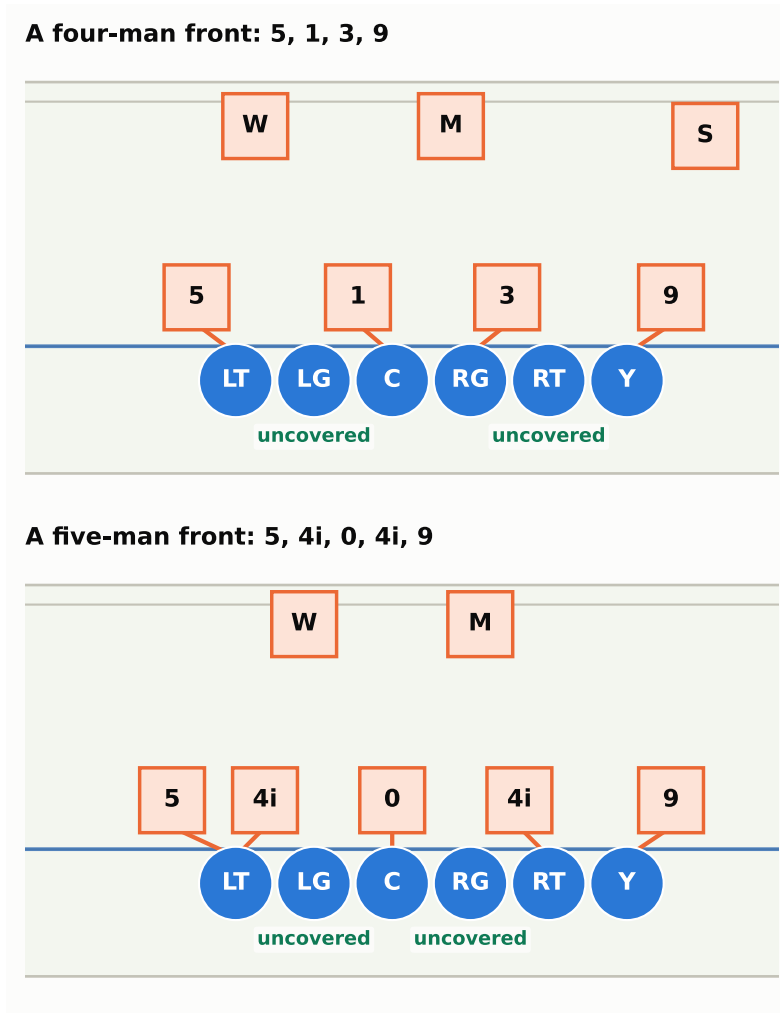


Figure 6: The same two fronts, now from the offensive line’s point of view. An orange tie links each defensive lineman to the offensive lineman he is lined up on, who is covered. Linemen with nobody on them (green) are uncovered: the nearest defender in front of them is a linebacker. Top: the LG and the RT are uncovered. Bottom: both guards are uncovered, the center and both tackles covered (the LT twice over).

An offensive lineman is **covered** when a defensive lineman is lined up on him, head-up or on either shoulder. He is **uncovered** when nobody is, so that the nearest defender in front of him is a linebacker four or five yards away. In the top front of Figure 6, the 5 covers the left tackle, the 1 covers the center, the 3 covers the right guard and the 9 covers the tight end; the left guard and the right tackle are uncovered.

Why every blocking scheme starts here. A covered lineman has a man in his face at the snap and usually has to deal with him first. An uncovered lineman is the offense’s free resource. He can help a neighbor on a defensive lineman (a double team), or go straight to a linebacker at the *second level*, the linebackers’ depth. Which uncovered linemen exist, and where, shapes what any run play can do against that front. Against the bottom front, both guards are uncovered and both have a linebacker in front of them, so the guards are the natural climbers; the center, covered by a 0, has the heaviest man on the field on his nose and little chance to help anyone. A defense that wants to stop a certain run chooses its front partly to change who is covered.

One refinement, which you will meet in Zone Running: zone-blocking teams count “covered” only on the side the play is going. A defender shaded on a lineman’s *backside* shoulder is not his problem on a zone run, because he is stepping away from him. That rule is why the same lineman can be covered on a run to the right and uncovered on a run to the left.

The blocks: eight words for what a blocker does

Run blocking looks like a pile of large men shoving. It is actually a small number of distinct jobs, each with a name, each chosen for a reason. This chapter teaches eight. They are enough to describe nearly every NFL run you will see; a few specialty blocks arrive later, in the chapters that first need them (the *cut block* in 03-02, the *log* in 03-03, and the *seal* and *crack* on the perimeter in 03-05).

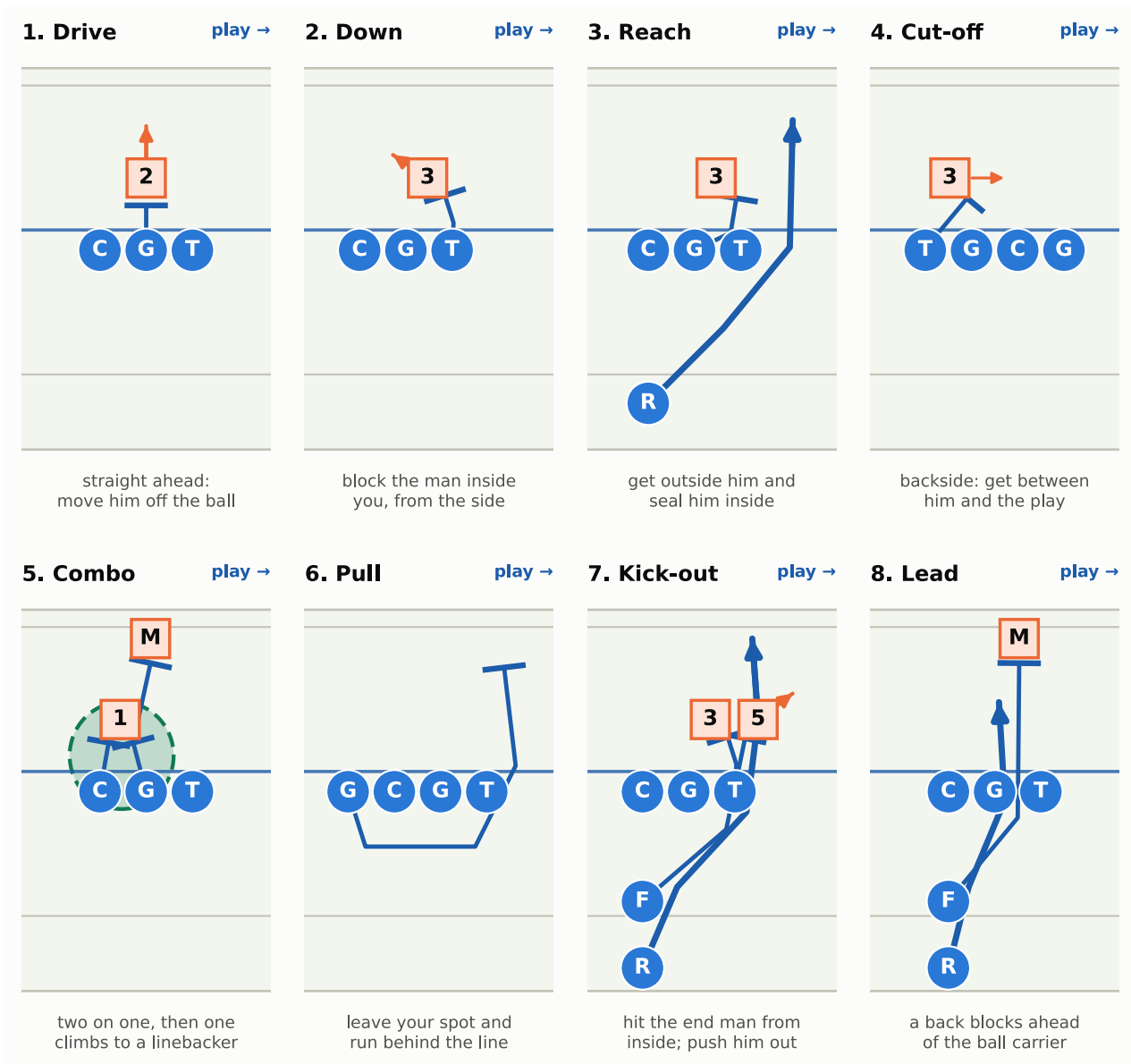


Figure 7: The eight core blocks, play going right in every panel. Defenders are drawn a little deeper than real (about a yard off the ball in a game) so each block's path shows; orange arrows show where the defender is pushed or wants to go. 1, Drive: straight ahead, move the man back. 2, Down: block a man lined up inside you, from the side. 3, Reach: get outside a man on your play-side shoulder and seal him in. 4, Cut-off: on the backside, step play-side and get between the defender and the play. 5, Combo: two linemen on one defender, then one climbs to the linebacker. 6, Pull: leave your spot and run behind the line. 7, Kick-out: hit the end man from the inside and push him toward the sideline; the run goes inside him. 8, Lead: a back runs through the hole ahead of the ball carrier and blocks the first defender who shows.

Work through Figure 7 one panel at a time.

1. Drive. The **drive block** is the simplest block in football: the defender is lined up on you, and you block him straight ahead and try to move him backward, off the ball. Every other block is a variation on it. *Why it exists:* moving a defensive lineman a yard or two backward collapses the space linebackers need to get to the hole. *What it costs:* it is a pure strength contest. Drive-blocking a 320-pound defensive tackle alone, head-on, is the hardest way to move him, and against a better player it simply fails.

2. Down. On a **down block** the blocker goes after a defender lined up *inside* him, toward the center, and hits him from the side. *Why it exists:* angles. A defender is built to take on force from the front; shove him from the side and he has much less to push back with, the way a parked car is easy to rock sideways and hard to push forward. A down block also *seals*: it pins the defender inside, building a wall on the inside edge of the hole. Gap-scheme runs are built out of down blocks. *What it costs:* the down-blocker leaves the man on *his* outside unblocked, so somebody else (a puller, a fullback) has to deal with that man.

3. Reach. A **reach block** is the opposite problem: the defender is on your *play-side* shoulder, a step closer to the play than you, and you have to get your head and body outside him so he cannot get to the ball, sealing him inside. *Why it exists:* runs to the outside need the defenders inside them to be kept inside. *What it costs:* it is the hardest block in football. The defender is already a step ahead and moving the way you want to go; he only has to take one step sideways to stay outside you. That is why the reach is usually done with help (the zone chapter shows the two-man version).

4. Cut-off. The **cut-off block** is a backside block. The blocker steps toward the play and gets his body between a defender and the ball carrier, usually a defender on his play-side shoulder, the one who could chase the play. *Why it exists:* backside defenders do not need to be moved, only kept from pursuing, and a body in their path for one second is enough. *What it costs:* little, which is the point; but if the defender beats it, he is the man who tackles a cutback runner from behind. (Do not confuse it with the *cut block*, a low block at the thighs, which is a different thing taught in 03-02.)

5. Combo. A **combo block**, also called a **double team**, puts two adjacent linemen on one defensive lineman: usually one covered and one uncovered. They drive him together, and both keep their eyes on the linebacker behind him. When the linebacker commits to a gap, one of the two comes off and **climbs** to the second level to block him; the other finishes the defensive lineman alone. *Why it exists:* it is the only way to get both things the run needs from one pair. Two men can move a defensive tackle off the ball when one man cannot, and the climb means the linebacker is still blocked. *What it costs:* timing. Come off too early and the defensive lineman splits the two; too late and the linebacker is already in the hole. The animation below shows it.

6. Pull. To **pull** is to leave your spot at the snap, step back off the line, and run sideways behind it to block somewhere else. It is mostly guards (and some centers and tackles) who pull. *Why it exists:* it adds a blocker where the play is going. A lineman lined up on the backside, where his block matters least, becomes an extra body at the point of attack, where it matters most. *What it costs:* the spot he left is now empty, so his neighbors must cover for it; pulling takes time, so a fast defender can beat the puller to the hole; and it is a *tell*. Linebackers are taught to read the guards, and a pulling guard tells them where the play is going. (Linebackers and the Second Level covers that read.)

7. Kick-out. A **kick-out** block hits the defense's *end man*, the last defender on the line on the play side, from the inside, and pushes him toward the sideline. The run then goes *inside* him. *Why it exists:* after the play-side linemen block down, the end man is the one defender left at the edge of the hole, and he is lined up outside everybody. Nobody on the line can reach him, so a blocker coming from the inside, a pulling guard or a fullback with a running start, takes him on with the angle in his favor. *What it costs:* an extra blocker, who has to come from somewhere (the backfield, or the backside of the line).

8. Lead. A **lead block** is thrown by a back, usually a fullback, who runs through the hole ahead of the ball carrier and blocks the first defender to show up, often a linebacker. *Why it exists:* the linemen can block the linemen, but somebody has to block the linebacker who is reading the play and running to the hole; the lead blocker arrives there at full speed, with the ball carrier right behind him. *What it costs:* a player who could otherwise run a route or carry the ball, which is one reason fullbacks became rare in the NFL. (The classic lead play, *iso*, is in 03-03.)

Notice a pattern in that list. Four blocks are about **winning a matchup** (drive, down, reach, cut-off): one blocker against one defender, with the block chosen to give the blocker the best angle. The other four are about **changing the numbers** (combo, pull, kick-out, lead): getting two blockers onto one defender, or getting a blocker to a spot where the defense did not expect one. That second group is where most run plays are won, and it is the subject of the rest of the chapter.

! Why it exists

Every block has a geometry. A defender is hard to move from the front and easy to move from the side, so offenses design runs so that as many blocks as possible are *angle* blocks (down blocks, kick-outs) rather than head-on drive blocks, and so that the hardest block, the reach, is done with help. When you watch a run and wonder why a lineman went *sideways* at the snap instead of straight ahead, the answer is almost always the angle.

The combo block, frame by frame

The combo is the block that confuses viewers most, because it changes halfway through. One moment two linemen are pushing the same man; a second later one of them is fifteen feet away on a linebacker. Here it is in slow motion. The animation (a numbered frame strip in the PDF) is a simulated play, drawn in the same coordinates as the NFL's tracking data, not a real snap.

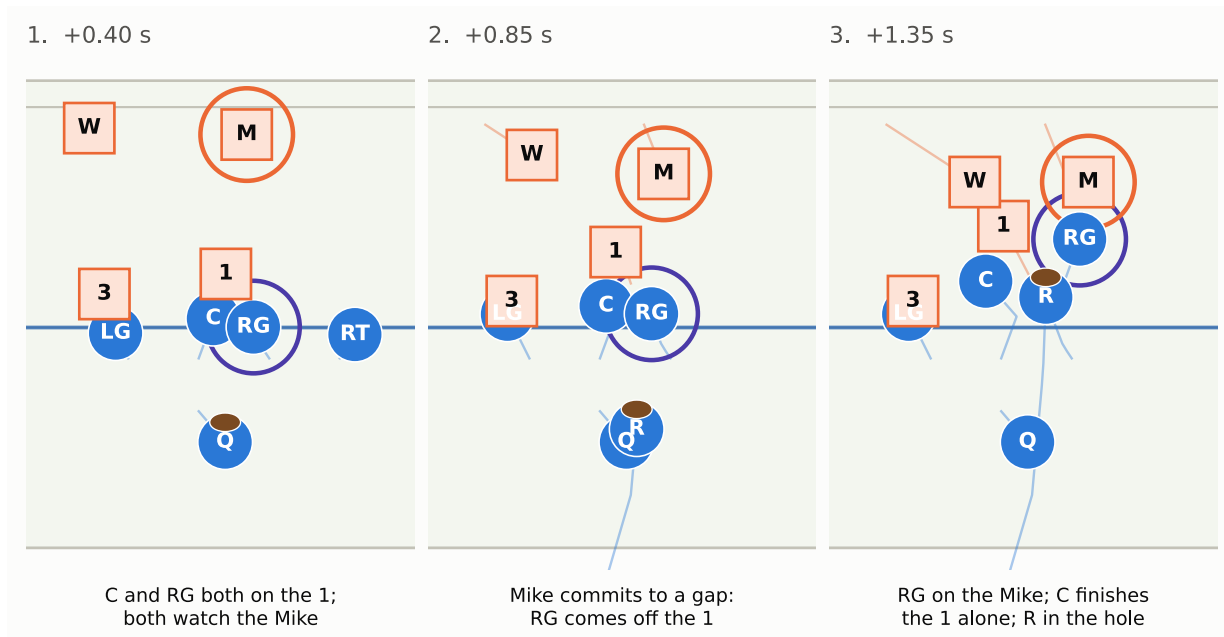


Figure 8: A combo block on an inside run to the right (simulated, in Big Data Bowl tracking coordinates; faint lines are each player's path so far). 1: the center and right guard (purple ring) both have their hands on the 1-technique, driving him back, and both are watching the Mike linebacker (orange ring). 2: the Mike reads run and steps up into a gap; that commitment is the guard's signal to come off the double team and climb. 3: the guard is on the Mike, the center has turned the 1-technique by himself, and the running back is in the hole the pair created.

Three things to notice in Figure 8. First, the 1-technique is moved *backward*, a yard or more, which one blocker alone could rarely do. Second, the guard does not leave until the Mike commits; he blocks the defensive lineman *and* reads the linebacker at the same time. Third, the hole appears in the space the double team cleared. Line coaches say the double team “creates the movement and the climb creates the hole,” and you will see it on nearly every inside run.

Against many modern fronts the climber is the center. When a four-man front puts both defensive tackles in 3-techniques (a common nickel look), nobody covers the center, so he is the free hat who helps a guard and then climbs. That is why centers who can *move*, who can come off a double team and catch a linebacker in open space, are so valuable in modern offenses. The Eagles' Jason Kelce, a sixth-round pick who weighed about 295 pounds, light for an NFL center, made six first-team All-Pro teams largely on that skill: climbing to linebackers and pulling around the edge faster than any center of his era.⁴

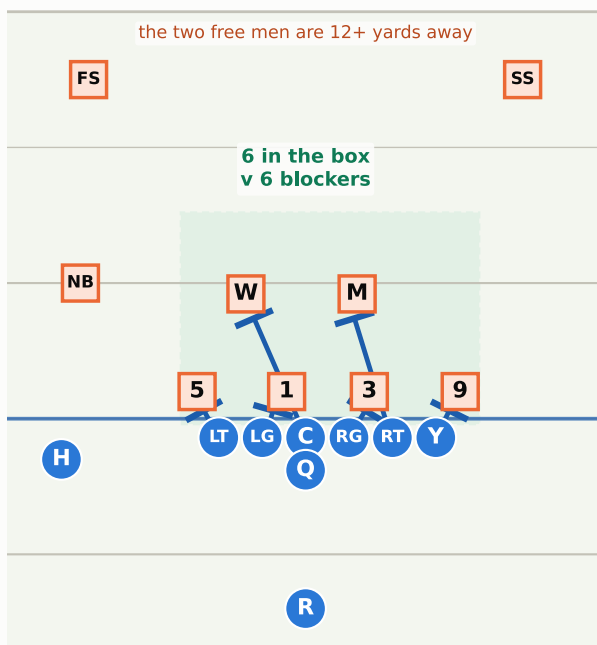
Hat math: counting blockers and defenders

Now the arithmetic. Coaches call it **hat math** (sometimes “counting hats,” a *hat* being a helmet, so a player): count the blockers, count the defenders, and see who is left over. The phrase you will hear most is “**a hat on a hat**”: every defender who can get to the run has a blocker assigned to him. If the defense has one more man than the offense has blockers, one defender is unblocked, and on a run that defender is usually the one who makes the tackle.

The count happens in **the box**, the area near the ball, roughly from tackle to tackle (or tight end to tight end) and five to seven yards deep, where defenders are close enough to play the run. Counting the box is taught in *A First Look at Defense*, along with *light* and *loaded* boxes; here is why that count matters so much to the run game.

⁴Jason Kelce: sixth round (191st overall), 2011; six first-team All-Pro selections (2017–2019, 2021–2023); listed at 6-3, 295; retired March 4, 2024: https://en.wikipedia.org/wiki/Jason_Kelce; Pro Football Reference: <https://www.pro-football-reference.com/players/K/KelcJa00.htm>.

Two deep: a hat on every hat



One deep: one man free in the box

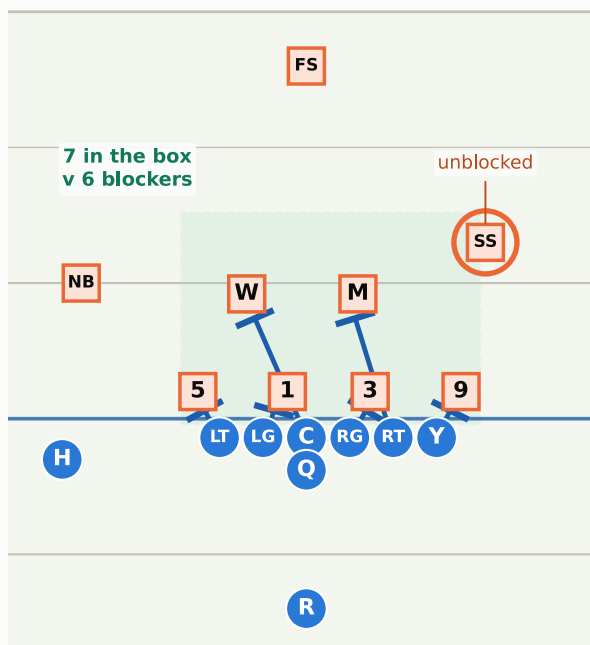


Figure 9: Hat math against one offense: one running back, one tight end, three receivers (11 personnel), so six blockers in the box (five linemen and the tight end). Left: the defense keeps two safeties deep, leaving six in the box (shaded): six blockers, six defenders, a hat on every hat. The two defenders nobody blocks are the safeties, more than 12 yards away. Right: one safety walks down into the box, making seven; the blocking scheme still has only six blockers, so one defender (ringed) is unblocked, five or six yards from the ball.

Read the two pictures in Figure 9 as a negotiation.

- **Six against six** (left). The offense has a blocker for every defender in the box. The run has a fair fight up front, and if it gets through the line, the next defenders are two safeties a dozen yards away. Offenses love to run against this look. It is common now, because defenses have leaned into two-deep safeties to stop the pass: NFL Next Gen Stats measured *split-safety* coverage on 42.0% of dropbacks in 2025, up from 32.9% in 2018.⁵
- **Seven against six** (right). The defense has rolled a safety down. Now there is one more defender than blockers, and the unblocked man is in the box, not twelve yards away. Some defender will be unblocked whatever the blocking scheme does; the scheme only gets to choose *which* one. A good run play leaves the defender who is farthest from the play, or the one the ball carrier can most easily make miss.

Why doesn't the defense just put everyone in the box? Because every man it adds there is a man taken out of pass coverage. In the right-hand picture there is now only one deep safety, and each receiver outside has a cornerback on him with less help over the top. That is the trade the box count measures. It is why quarterbacks count the box before the snap and switch from a run to a pass when the numbers are bad (the *kill call* in The Pre-Snap Phase), and why defenses try to disguise how many men will really be in the box at the snap. Defensive coaches have a name for having one more than the offense can block: being plus-one in the run game, which is the subject of 05-05.

The offense can add blockers, of course. Bring in a second tight end or a fullback and there are seven blockers. But every blocker you add is a receiver you took off the field, so the defense needs fewer men in coverage and can bring another one into the box. The defense can match each added blocker with an added defender, one for one, and keep its extra man. The next section explains why that extra man is almost always there.

Why a quarterback who only hands off leaves you a man short

Count everyone on the field, not just the box. Eleven offensive players, eleven defensive players. On a normal handoff:

- The **ball carrier** does not block anyone; he is the man the defense is trying to tackle.

⁵NFL Next Gen Stats: split-safety coverage on 32.9% of dropbacks in 2018 and 42.0% in 2025, the highest since tracking began: <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm>.

- The **quarterback** hands the ball off and then, usually, does nothing that matters. Nobody has to cover him or tackle him. For the rest of the play he is a spectator.
- The other **nine** players (five linemen, and four tight ends, backs and receivers) each block a defender or take one away by running a route that a defender has to cover.

So nine offensive players occupy nine defenders, and **two defenders are left free** with nobody assigned to them. The defense gets to choose where those two free men stand. Put both deep (the two-high picture) and the run gets a fair fight in the box but runs into two safeties if it breaks through. Put one in the box (the one-high picture) and that man can tackle the runner near the line. Either way, the offense is playing ten men against eleven. The quarterback is the missing man.

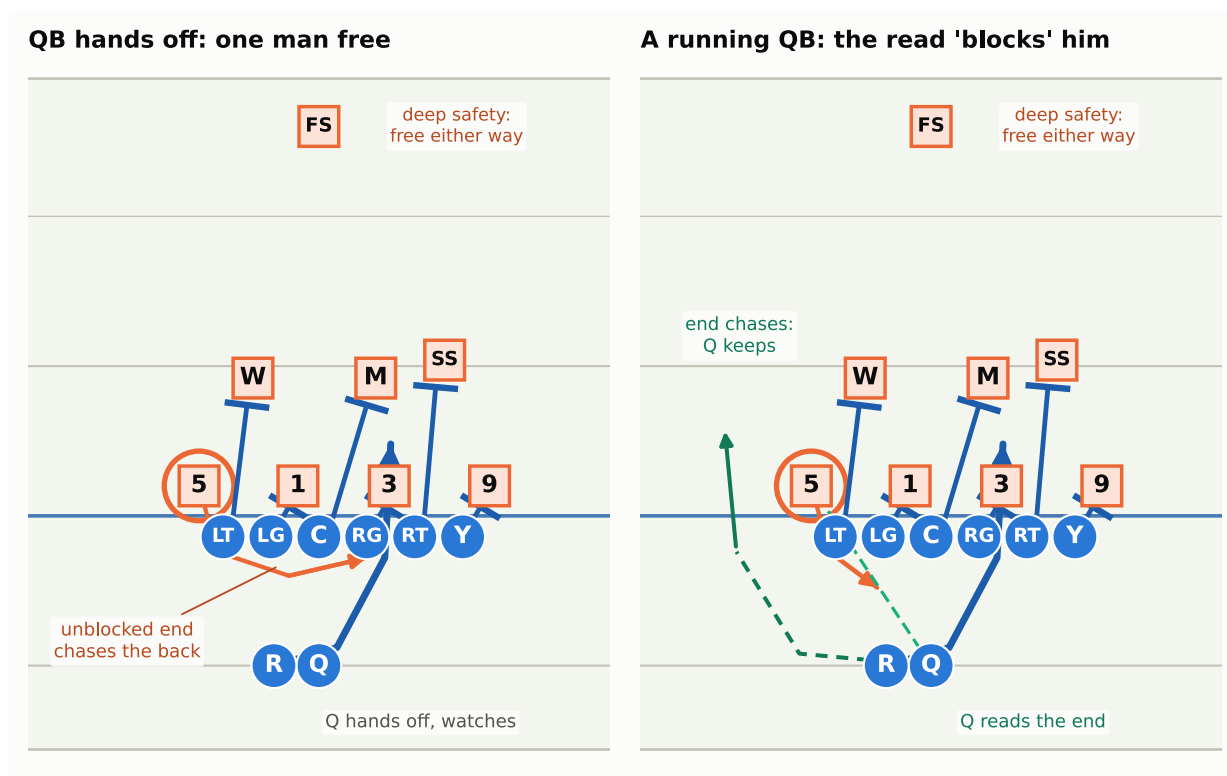


Figure 10: The defender a handoff leaves free, from the shotgun against seven in the box. Left: the quarterback hands off and watches. Six blockers take six defenders, and the offense chooses to leave the backside end (ringed) alone; nothing stops him chasing the running back down the line from behind. Right: the same play, but the quarterback reads that end (green dashes). If the end chases the back, the quarterback keeps the ball and runs where the end just left. The end can no longer chase freely, because the quarterback has ‘blocked’ him without touching him. The deep safety is free either way.

Now give the quarterback a job. If he is a real threat to run, a defender has to account for him, and that defender cannot also tackle the running back. The cleanest version is the **read**: the offense deliberately leaves one defender unblocked and the quarterback watches him after the snap. If that defender plays the running back, the quarterback keeps the ball; if he plays the quarterback, the quarterback hands off. Either way the read has done a blocker’s work: the defender is “blocked” by the decision. The offense is back to eleven against eleven, and one of the defense’s free men has disappeared. That is the core idea of the zone read and every other option play, which Option Football builds out properly. Jim Harbaugh, whose 49ers ran the read option with Colin Kaepernick, put it plainly in the week before Super Bowl XLVII, crediting the thought to Lou Holtz: without an option attack “the offense is basically playing with 10,” but when the quarterback is a threat to run, “that gets the numbers back to even, back to 11-on-11.”⁶

That arithmetic is why running quarterbacks change everything about a run game. In 2019 the Baltimore Ravens, with Lamar Jackson running a read-option-heavy offense, ran for 3,296 yards, breaking a team record that had stood since 1978; Jackson’s own

⁶Kareem Copeland, “Jim Harbaugh: Option allows 49ers to play 11-on-11,” NFL.com, January 31, 2013: <https://www.nfl.com/news/jim-harbaugh-option-allows-49ers-to-play-11-on-11-0ap1000000133218>. Chris B. Brown’s *The Essential Smart Football* (2012) and *The Art of Smart Football* (2015) develop the same arithmetic.

⁷“Ravens break single-season team rushing record,” NFL.com (Dec 2019; 3,296 yards, breaking the 1978 Patriots’ 3,165): <https://www.nfl.com/news/ravens-break-single-season-team-rushing-record-0ap30000001092761>; Baltimore Ravens, “Ravens set all-time record for team rushing yards in a season”:

1,206 yards were a record for a quarterback.⁷ The quarterback did not have to run every time. He only had to be a threat that someone had to watch.

There is a quieter version of the same idea that every offense uses: the **pass threat**. Each receiver who might run a route takes a defender with him, which is why defenses facing three or four receivers keep fewer men in the box. A quarterback who can make the defense pay for loading the box with an accurate throw is, in run-game arithmetic, worth a blocker too. And when the quarterback runs the ball himself on a designed run (a sneak, or the *QB power* of 03-04), the ball carrier is the quarterback, so the running back becomes a blocker: the offense gains a hat. That is the arithmetic behind the tush push, too (Red Zone, Goal Line, and Short Yardage).

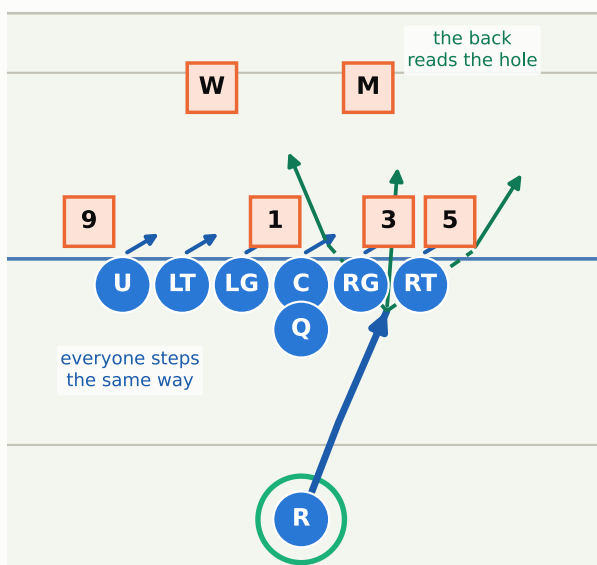
🔥 Common misconception

“Running quarterbacks are dangerous because they are fast.” Speed helps, but the bigger effect is arithmetic. A quarterback the defense has to account for on every run takes a defender away from the running back, even on the plays where he hands off. That is why an offense can get a better run game from a quarterback who keeps the ball only a few times a game: the threat does the work.

Two philosophies: zone and gap

Every running play in this book is built from the pieces above, and coaches assemble them in two great families. You will spend a chapter on each; here is the map.

Zone: block an area



Gap: block a man, add a blocker

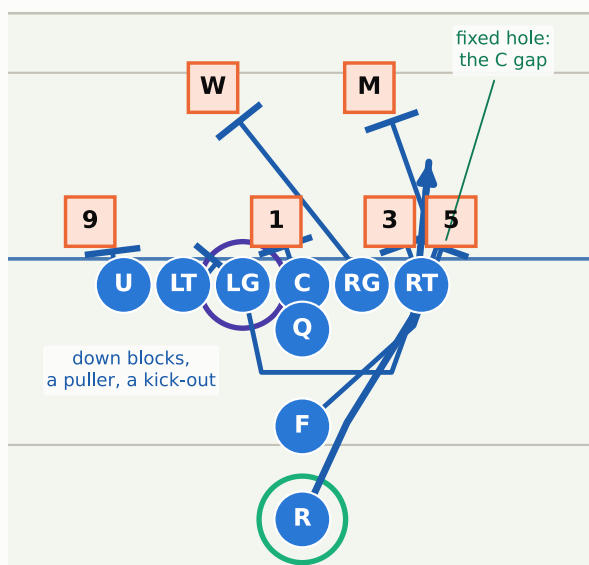


Figure 11: The two blocking philosophies against the same four-man front, both aimed to the right. Left, zone: every lineman’s first step goes the same way (short arrows), each blocks whoever comes into his area, and the running back reads the blocks and picks one of several possible lanes (green dashes). Right, gap (this is the play called power): the play-side linemen block down on the men inside them, the center blocks back, the left guard (purple ring) pulls around to lead up through the hole on the Mike, and the fullback kicks out the end man. The hole is fixed in advance: the C gap, inside the kick-out.

Zone blocking assigns blockers to *areas*. All the linemen step the same direction at the snap, each takes whoever enters his area, uncovered men double-team and climb, and the running back reads where the wall cracks. Its virtue is that it adapts: if the defense slants or blitzes, whoever arrives gets blocked. Its price is that blocks are rarely knock-down blocks, and the hole is not known in advance, so the play depends on a runner who can see it. (Zone Running is the full story.)

<https://www.baltimoreravens.com/news/ravens-set-all-time-record-for-team-rushing-yards-in-a-season>; Jackson’s 1,206 yards: https://en.wikipedia.org/wiki/2019_Baltimore_Ravens_season.

Gap blocking (also called *man* or *power* blocking) assigns blockers to *men*, chosen to give them angles, and moves extra blockers to a hole chosen in advance. Down blocks seal the inside, a kick-out walls off the outside, and a puller or a fullback leads through the space between them. Its virtue is numbers at the point of attack: it gets more blockers to the hole than the defense has defenders there. Its price is that it is easier to read (a pulling guard is a signpost) and more fragile against movement: if a defender slants into the space a puller was supposed to run through, the play can die. (Gap Schemes covers power, counter, trap and their relatives.)

Neither is better. Almost every NFL team runs both, and good offenses build them to look alike for the first half-second so the defense cannot tell which is coming. The first thing to watch on any run is which family it belongs to, and you can usually tell from one question: **does the whole line step the same way, or does somebody pull?**

Where this language came from: the Hogs

Most of this vocabulary is older than anyone now coaching, but one team made it famous to a national audience, and its story explains why the blocks in this chapter are still the foundation.

When Joe Gibbs became Washington's head coach in 1981, his offense was built to run behind a big offensive line. Line coach **Joe Bugel** started calling his linemen "the Hogs" in training camp in 1982 ("Okay, you hogs, let's get running down there"), and the name stuck to a group of big, unglamorous blockers (center Jeff Bostic, guards Russ Grimm and Mark May, tackles Joe Jacoby and George Starke, among others) who became the most famous offensive line in football.⁸ Washington won three Super Bowls under Gibbs, after the 1982, 1987 and 1991 seasons, with three different starting quarterbacks, which tells you how much of the offense lived up front.

Gibbs's teams are a good place to see the language in use because their signature runs were the gap-scheme plays this chapter just described: down blocks on the play side, guards and even tackles pulling around to kick out and lead, a fixed hole. Their counter play, with two backside linemen pulling (the left guard Russ Grimm and, unusually, the left tackle Joe Jacoby), became one of the most copied runs in football (it is taught in 03-03).⁹ And Gibbs's answer to one defender, the Giants' Lawrence Taylor, gave the run game a new blocker: Gibbs moved a back up near the line as an extra blocker for the edge, the position now called the *H-back*.¹⁰ That is hat math again. Taylor was a defender the existing blocking could not handle, so Gibbs added a hat.

i Film room: Washington, Super Bowl XVII (1982 season)

The play: January 30, 1983, at the Rose Bowl. Washington trails Miami 17–13 with 10:10 left in the game and faces fourth-and-1 at the Miami 43. Gibbs calls "I-Right 70 Chip," a short-yardage run to the left. Tight end Clint Didier goes in motion to the right and then reverses back to the left, and cornerback Don McNeal follows him. The play is designed, as the standard account of it puts it, to block everyone except the smallest defender on the field, the cornerback, and let John Riggins run through him. Behind Didier and fullback Otis Wonsley, Riggins breaks McNeal's tackle at the line (McNeal had slipped as Didier changed direction) and outruns safety Glenn Blackwood for a 43-yard touchdown. Washington wins 27–17; Riggins carries 38 times for 166 yards and is named MVP.¹¹

What to notice: this is hat math in its purest form. The offense could not block everybody, so it chose the defender to leave free: the smallest one, against a 230-pound back with a running start. Watch the replay with the vocabulary: which side is the play side, which gap Riggins hits, who leads (the fullback), and what the motion did to the cornerback before the snap. (Motion and Shifts is about that last idea.)

Five years later the same line produced an even more lopsided day: in Super Bowl XXII, after the 1987 season, rookie Timmy Smith,

⁸"The Hogs (American football)," Wikipedia (Bugel coined the name in 1982 training camp; original members; three Super Bowl titles): [https://en.wikipedia.org/wiki/The_Hogs_\(American_football\)](https://en.wikipedia.org/wiki/The_Hogs_(American_football)). Russ Grimm is in the Pro Football Hall of Fame (class of 2010): <https://www.profootballhof.com/players/russ-grimm/>.

⁹Rick Gosselin, "State Your Case: Joe Jacoby" (2015; updated 2022): on the counter Trey "the left guard and left tackle pull to the right," and Gibbs: "You can pull guards. But how many people pull tackles? We pulled Joe. He could run." <https://rickgosselin.com/state-your-case-joe-jacoby/>.

¹⁰"H-back," Wikipedia (Gibbs's one-back offense and the H-back as a response to Lawrence Taylor, with Gibbs's own account): <https://en.wikipedia.org/wiki/H-back>. See also the Pro Football Hall of Fame on Taylor: <https://www.profootballhof.com/news/2023/06/gold-jacket-spotlight-lawrence-taylor-created-havoc/> (Gibbs: "You get a back blocking Lawrence Taylor, you lose"). Gibbs's own account, as cited by Wikipedia: Nick Charles, "Taylor made," CNN.com, August 12, 1999 ("We had to try in some way have a special game plan just for Lawrence Taylor").

¹¹"Super Bowl XVII," Wikipedia (fourth-and-1 at the Miami 43 with 10:10 left in the fourth quarter, trailing 17–13; "I-Right 70 Chip"; Didier's motion and McNeal's slip; the design to block everyone but the cornerback; the run to the left behind Didier and Otis Wonsley; Riggins 38 carries, 166 yards, MVP; Washington 27, Miami 17): https://en.wikipedia.org/wiki/Super_Bowl_XVII; Riggins listed at 6-2, 230 lb: <https://www.profootballhof.com/players/john-riggins/>; Pro Football Reference box score: <https://www.pro-football-reference.com/boxscores/198301300mia.htm>.

making his first career start, ran for 204 yards, still the Super Bowl record, in a 42–10 win over Denver.¹²

The modern echo of the Hogs is not one team but one position. As offenses spread out and defenses got faster, the center became the hub of the run game: the man who makes the line's calls, who is usually uncovered and must climb, and who in some offenses pulls. Two recent lines show the vocabulary at its best.

i Film room: Philadelphia Eagles 2022

The season: the 2022 Eagles went 14–3 and reached Super Bowl LVII, which they lost 38–35 to Kansas City. By this book's calculation from nflverse play-by-play, they were first in the league in both EPA per designed run and success rate on designed runs, and first in rushing touchdowns on designed runs. Those numbers include Jalen Hurts's designed runs and the Eagles' short-yardage quarterback sneaks, a reminder from the last section that a quarterback who runs is part of the blocking arithmetic.¹³

What to notice: watch center **Jason Kelce** (#62) on every run for a quarter. Count how often he is the lineman who comes off a double team and climbs to a linebacker, and how often he pulls or sprints out in front of a play to the edge. Then watch his guards: on most of the plays where Kelce climbs, a guard is finishing the defensive lineman Kelce left. That is the combo block, done at a speed most centers cannot match.

i Film room: Detroit Lions 2023–2024

The seasons: under offensive coordinator Ben Johnson, Detroit reached the NFC Championship Game after the 2023 season and went 15–2 in 2024. By the same nflverse calculation, the Lions were in the top five in EPA per designed run in both seasons and tied for the league lead in rushing touchdowns on designed runs both years.¹⁴ Their line was anchored by right tackle **Penei Sewell** (#58), a first-team All-Pro in 2023, 2024 and 2025, and center **Frank Ragnow** (#77), a second-team All-Pro in 2023 and 2024 who retired in June 2025 (a comeback attempt that November ended when he failed his physical).¹⁵ (Johnson became the Bears' head coach in 2025, so this is a 2023–2024 offense, not Detroit's current one.)

What to notice: pick a game and watch only the right side. Sewell's blocks are a block-library tour: down blocks on runs to his side, drive blocks that move defensive ends a couple of yards off the ball, reach and cut-off blocks when the play goes away from him. Then watch Ragnow on the double teams in the middle: who climbs, and when. Name each block out loud as it happens.

¹²"Timmy Smith," Wikipedia (204 yards in Super Bowl XXII, January 31, 1988, in his first career start; Washington 42, Denver 10): https://en.wikipedia.org/wiki/Timmy_Smith; Guinness World Records, "Most yards gained rushing in a Super Bowl game": <https://www.guinnessworldrecords.com/world-records/75399-most-yards-gained-rushing-in-a-super-bowl-game-amf>.

¹³This book's calculation from nflverse play-by-play (`load_pbp`), 2022 regular season: plays with `play_type == "run"`, excluding quarterback scrambles and kneels, so quarterback designed runs and sneaks are included. Philadelphia ranked 1st of 32 in EPA per play (+0.09) and success rate (51%) and had 30 touchdowns on these plays. Season record and Super Bowl LVII result: Pro Football Reference, <https://www.pro-football-reference.com/teams/phi/2022.htm>.

¹⁴Same calculation as the Eagles note, 2023 and 2024 regular seasons: Detroit ranked 5th in EPA per designed run both years and tied for the most designed-run touchdowns both years (27 in 2023, with San Francisco and Miami; 29 in 2024, with Philadelphia and Buffalo); 5th in success rate in 2024, 8th in 2023. Ben Johnson was Detroit's offensive coordinator 2022–2024 and became Chicago's head coach for 2025 (this book's checked facts, planning/FACTS-current.md §9).

¹⁵Penei Sewell (first-team All-Pro 2023, 2024, 2025): https://en.wikipedia.org/wiki/Penei_Sewell. Frank Ragnow (second-team All-Pro 2020, 2023, 2024; retired June 2, 2025; announced a return on November 26, 2025, which ended three days later when he failed his physical with a hamstring strain): https://en.wikipedia.org/wiki/Frank_Ragnow. Sewell's third straight first-team selection (2025 team, announced January 2026): <https://www.foxsports.com/articles/nfl/2025-nfl-allpro-team-see-who-made-the-roster>.

Watch for it on Sunday

Watch for it

On Sunday: find the hole. For one quarter, on every run, do this.

Before the snap

- Count the box and the blockers (five linemen, plus each tight end and fullback near the line). Six against six? Seven against six? Who would be the free man?
- Read the defensive line in numbers, left to right (“5, 1, 3, 9”). Which guard has a 3-technique on him? Which linemen are uncovered?

The first half-second

- Watch the running back’s **first step**. It points at the play side and usually at the point of attack. Say the gap out loud before he gets there: “right B gap.”
- Glance at the guards. Does one pull? (Gap play, and the puller is running to the hole.) Does the whole line step together? (Zone, and the hole is not decided yet.)

After the play (on the replay)

- Name the gap he actually hit. Same as your guess? On a gap play it usually is; on a zone play it often is not, and that is the play working, not failing.
- Find one block of each kind you can see: a down block, a double team with a climb, a kick-out, a lead.
- Find the unblocked defender. Where was he, and did he make the tackle?

For your prediction log: before each run write the gap you expect (say, “R-B”); afterwards write the gap he hit and Z (zone) or G (gap). After a quarter you will know whether this offense’s first step tells the truth.

Predict the play

Predict the play

Drill 1. Find the hole. First-and-10. Figure 12 shows the first half-second of a run: the linemen's first steps, a pulling guard (purple ring), the fullback heading for the edge, and the running back's first two steps. The defense has a linebacker (S) standing up on the line outside the tight end. Where is the point of attack (name the gap)? Name the block the Y and the RT are throwing, and say what the fullback and the pulling guard are about to do.

Drill 1: the first half-second. Where is it going?

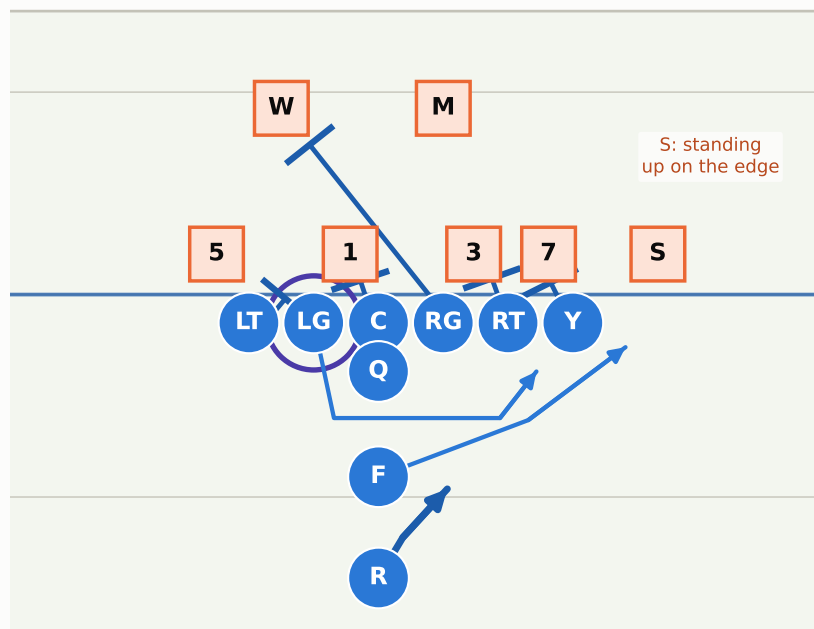


Figure 12: Drill 1: the first half-second of a run from the I-formation. The tight end and right tackle have stepped inside, the left guard (purple ring) is pulling to the right, the fullback is heading outside, and the running back has taken two steps to the right.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 2. Count the hats. Second-and-1. The offense brings in a fullback and lines up in the I-formation with one tight end (21 personnel); the defense answers with four down linemen, three linebackers and two deep safeties. Do the hat math for a run. Who is unblocked? If the defense wants to stop the run harder, what will it do, and what does that give the offense?

Drill 2: 2nd and 1, 21 personnel, a base defense

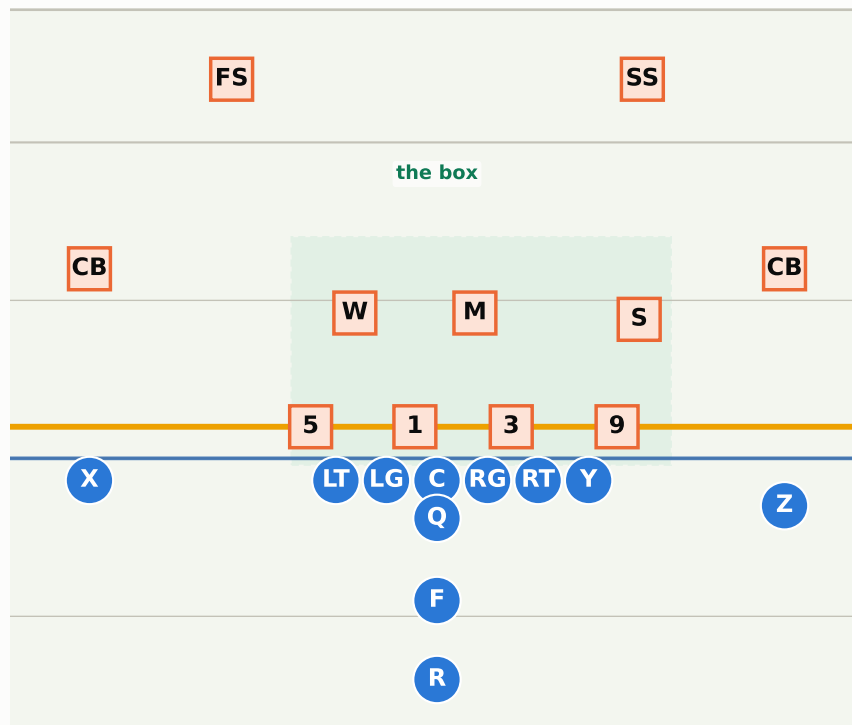


Figure 13: Drill 2: second-and-1 (yellow line one yard ahead). 21 personnel in the I-formation against a base defense with two deep safeties. The shaded area is the box.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 3. Read the front. The defense's linemen are drawn with question marks. Read their alignments as technique numbers, left to right. Then: which offensive linemen are covered and which are uncovered? If the offense runs an inside run to the right, where will its double teams most naturally form, and who will climb to the linebackers?

Drill 3: name the front, then find the double teams

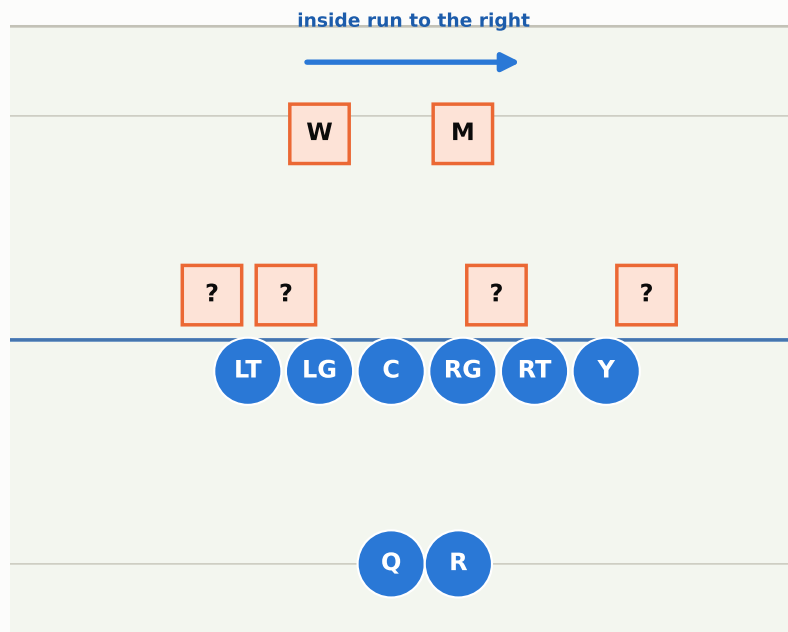


Figure 14: Drill 3: shotgun, one tight end on the right. Four defensive linemen (marked ?) and two linebackers. The offense will run inside, to the right.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Gaps and holes name the spaces.** Gaps are lettered A, B, C, D outward from the center (the defense's language; every gap needs a defender, and every blocker the offense attaches to its line adds one). Holes are the offense's numbers, usually odd on one side and even on the other, and "24" means the 2-back through the 4 hole, though numbering is a dialect.
- **Technique numbers say where a defender stands:** even numbers head-up (0 center, 2 guard, 4 tackle, 6 tight end), odd numbers on an outside shoulder (1, 3, 5, and 9 outside the tight end), "i" on an inside shoulder (2i, 4i), and 7 inside the tight end. A shaded defender usually owns the gap he is shaded into; the **3-technique**, **4i** and **5-technique** have become names for positions.
- **Covered or uncovered** is the first question every lineman asks. Covered linemen have a man; uncovered linemen are the free hats, the ones who double-team and climb.
- **Eight blocks describe almost every run.** Four win a matchup with the best angle available (drive, down, reach, cut-off); four change the numbers (combo, pull, kick-out, lead). Play-side blocks make the hole; backside blocks stop pursuit.
- **Hat math decides runs before the snap.** A handoff leaves the quarterback a spectator, so the offense plays ten against eleven and the defense has two free men to place, deep or in the box. A quarterback who can run, or a read that "blocks" a defender without touching him, gets one back.
- **Zone and gap are the two answers.** Zone blocks areas and lets the runner find the hole; gap blocks men with angles and brings extra blockers to a hole chosen in advance. Watch the first step: together, or does someone pull?

Connections

Builds on: The Twenty-Two (the positions, the nose tackle, the Mike, Will and Sam), The Pre-Snap Phase (counting the box and the kill call), Classic Formations (the I-formation and singleback sets in these diagrams) and A First Look at Defense (the box, one-high and two-high shells).

Sets up:

- Zone Running: the zone family in full, including how zone teams count “covered” by play side, the two-man reach, and the cut block.
- Gap Schemes: power, counter, duo, trap, iso and the draw, built from down blocks, pullers, kick-outs and lead blocks.
- Option Football: the read that “blocks” a defender, and the quarterback as the extra hat.
- Pass Protection: the same covered-and-uncovered counting, applied to rushers.
- Defensive Line Play: technique numbers from the defense’s side, one-gap and two-gap play, and how defensive linemen fight each block in this chapter.
- Run Fits and Box Math: the defense’s half of hat math.
- Kickoffs and Returns: a return is a run play, and its blocks use this vocabulary.

See also (case study): The Eagles: Trenches and the Tush Push; for the running-quarterback arithmetic at its peak, The Ravens’ Lamar Jackson Run Game.

If you want to go deeper: Chris B. Brown’s *The Art of Smart Football* and *The Essential Smart Football* (the quarterback run game and the arithmetic of the box); Pro Football Focus’s “Defensive Line Techniques: The Prototypes” for the numbering with player examples; and any Washington game from the early 1980s, watched for nothing but who blocks down and who pulls.

Go deeper

Hat math in the data, and why it is slippery. The NFL’s participation data, which nflverse publishes, records how many defenders were in the box at the snap. You might expect runs into seven-man boxes to do clearly worse than runs into six-man boxes. Here is what 2023–2025 shows for one personnel group in one situation.

Defenders in the box	Plays	Run share	Runs	Success rate	Yards per carry
5	1,145	17%	193	39%	4.9
6	6,523	47%	3,043	38%	4.9
7	2,030	55%	1,115	36%	4.8
8	247	71%	175	42%	5.6

Runs and box counts, 2023–2025 regular seasons: 11 personnel (one back, one tight end, five offensive linemen), first-and-10 between the 20s, win probability 20–80%. Box count is from nflverse participation data (FTN charting); “run share” counts designed runs (no scrambles or kneels) among runs and passes; success means positive EPA, and quarterback sneaks are left out of the run columns.

Two things jump out. Offenses do respond to the count: they ran on a small share of snaps against light boxes, which usually come with more defenders deep and wide. But the runs that offenses *did* call into seven- and eight-man boxes did not fare dramatically worse, and the share of plays that are runs goes *up* as the box gets heavier. That is not hat math failing. It is selection. Box counts are not random: defenses load the box when the offense’s formation and tendencies say run, and offenses that see seven in the box often check out of the weakest runs, so the runs that survive are the ones the offense still liked. A simple comparison of averages cannot separate the arithmetic from the decisions made around it. That problem, and the tools for it, are in Expected Points, EPA and Success Rate and Tendencies.

From diagrams to real tracking data. The combo-block animation is a gridiron simulation written in NFL Big Data Bowl coordinates, so real tracking data drops into the same drawing code. The 2025 Big Data Bowl release (2022 season, weeks 1–9) includes every player’s position ten times a second, before and after the snap, and its `plays.csv` carries PFF’s run-concept labels, so you can find inside runs and watch a center and guard come off a double team for real. The data needs a free Kaggle login, so the code (in the web edition, under “Show the Python”) is not run when this book is built.

Answers to Predict the play

Answer 1.

The D gap, just outside where the tight end lined up. Everything on the play side is going *inside*: the Y is down-blocking the 7-technique and the RT is down-blocking the 3-technique, hitting each from the side and sealing the inside of the hole. That leaves the end man, the S standing up outside the tight end, for the **fullback to kick out**: he will hit him from the inside and push him toward the sideline. The **pulling left guard** turns up into the space between the down blocks and the kick-out and **leads** through it onto the first linebacker who shows (here the Mike, flowing over). The running back follows the puller. Inside the hole: down blocks. Outside it: the kick-out. Through it: a lead. That combination is the play called *power*, the signature gap-scheme run, and 03-03 builds it fully.

The tell you could have used: the back’s first steps aimed outside the tackle, and the guard pulled the same way. A zone run does not pull guards; a pulling guard almost always runs to the hole.

Answer 2.

Seven against seven: a hat on every hat. In the box are seven defenders: four linemen (a 5, a 1, a 3 and a 9) and three linebackers. The offense has seven blockers: five linemen, the tight end, and the fullback as a lead blocker. Every defender in the box can be blocked, and the two free defenders are the safeties, twelve yards deep. On second-and-1 that is a very good look to run into: the runner needs one yard, and nobody unblocked is within ten of him.

If the defense wants the run more, it brings a safety down into the box: eight against seven, plus-one again. But that leaves one deep safety for two receivers and a tight end, and on second-and-1 the offense can afford a shot: play-action, with the fullback and tight end selling the run, against a single safety. This is the back-and-forth of hat math: the offense adds a blocker, the defense adds a defender, and whoever over-commits gets punished by the other half of the playbook.

Answer 3.

5, 3, 3, 9. From the left: a 5-technique on the left tackle’s outside shoulder, a 3-technique on the left guard’s outside shoulder, a 3-technique on the right guard’s outside shoulder, and a 9 outside the tight end. (Two 3-techniques is a common modern nickel look: both defensive tackles in the B gaps, ready to penetrate.)

Covered: LT (by the 5), LG (by the left 3), RG (by the right 3), Y (by the 9). **Uncovered:** the center and the right tackle. Those two are the free hats. On an inside run to the right, the natural pair is the **right guard and right tackle on the right 3-technique**: the RG is covered, the RT is uncovered beside him, so they double-team the 3 and the **RT climbs** to the Mike. The **center**, uncovered with nobody on him, can help the left guard on the other 3-technique and then climb to the Will. The Y has the 9 by himself, and on an inside run to the right that is the edge of the play.

Notice what the front did: by putting both tackles in 3-techniques, the defense left the center uncovered, so the center becomes the climber, which is why NFL offenses prize centers who can move.